

Matrices

4.8 Eigenvalues and Eigenvectors

Square matrix $A_{n \times n} X_{n \times 1} = \lambda X_{n \times 1}$, Where λ is eigenvalue.

$$\Rightarrow (A - \lambda I)X = 0$$

Taking determinant $\Rightarrow |A - \lambda I|X = 0$

Now, $f(\lambda) = 0 \Rightarrow$ The Characteristic equation

Example: If A is 3×3 matrix, $f(\lambda)$ will be a cubic equation.

$\lambda_1, \lambda_2, \lambda_3$ are eigenvalues.

$\Rightarrow$ Every matrix follows its own characteristic equation

And $X_1, X_2, X_3 \rightarrow$ Eigenvectors corresponds to $\lambda_1, \lambda_2, \lambda_3$.

Example: If $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$, Then what will be eigenvalues, inverse, and eigenvectors of the given matrix A?

Solution: For eigenvalues:

$$[A - \lambda I] = 0 \Rightarrow \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} -5-\lambda & 2-0 \\ 2-0 & -2-\lambda \end{bmatrix} = 0$$

$$(-5-\lambda)(-2-\lambda) - 4 = 0 \Rightarrow \lambda^2 + 7\lambda + 6 = 0$$

$$\Rightarrow \lambda_1 = -6 \text{ \& } \lambda_2 = -1 \text{ are the eigenvalues.}$$

By Cayley Hamilton Theorem, $A^2 + 7A + 6I = 0$

$$\Rightarrow A^2 = -[7A + 6I]$$

For Inverse:

$$A^{-1}(A^2 + 7A + 6I) = 0 \Rightarrow A + 7I + 6A^{-1} = 0 \Rightarrow A^{-1} = \frac{-1}{6}[A + 7I]$$

$$A^{-1} = -\frac{1}{6} \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}$$

Or

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{\det A} \begin{bmatrix} a_{22} & -a_{21} \\ -a_{12} & a_{11} \end{bmatrix}$$

$$\det A = 10 - 4 = 6 \Rightarrow A^{-1} = \frac{1}{6} \begin{bmatrix} -2 & -2 \\ -2 & -5 \end{bmatrix}$$

For Eigenvector:

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \Rightarrow \lambda_1 = -6 \text{ \& } \lambda_2 = -1$$

Eigenvector:

For $\lambda_1 = -6$:

$$[A - \lambda I]X = 0 \Rightarrow [A + 6I]X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x_1 + 2x_2 = 0, \quad 2x_1 + 4x_2 = 0$$

$$\Rightarrow x_1 = -2x_2$$

Let $x_2 = 1$, Then $x_1 = -2$

$$\text{So } X_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \Rightarrow \text{Normalized form: } X_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

For $\lambda_2 = -1$:

$$[A + I]Y = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-4y_1 + 2y_2 = 0 \quad \& \quad 2y_1 - y_2 = 0 \Rightarrow y_2 = 2y_1$$

Let $y_1 = 1$, Then $y_2 = 2$

$$\text{So } X_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \Rightarrow \text{Normalized form: } X_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\text{Now } A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \because A^T = A \Rightarrow \text{Symmetric Matrix}$$

$$X_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}, X_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \& \quad X_1^T = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \end{bmatrix}$$

$$X_1^T X_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2 \times 1} = \frac{1}{5} [-2 + 2] = 0$$

So for a symmetric matrix, the eigenvectors are orthogonal.

Location of Eigenvalues

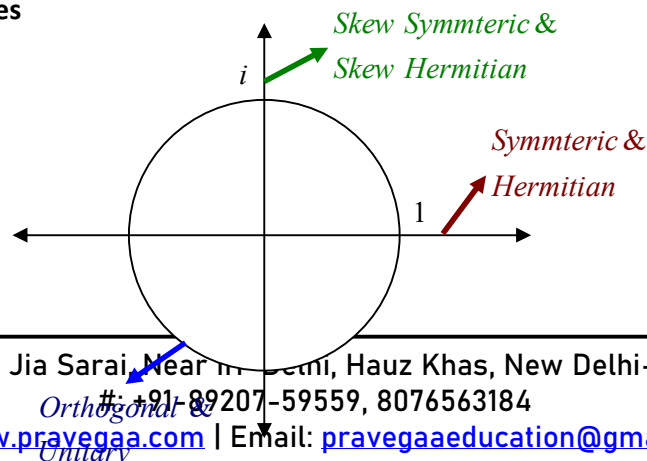


Fig. Argand Plane

Question: $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$, Find the Eigenvalues, Determinant, Trace, and Eigenvector

corresponding to eigenvalues.

Solution: $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{vmatrix} = 0$

$$\Rightarrow \lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

$$(\lambda + 3)^2 (\lambda - 5) = 0, \text{ Eigenvalues are } \lambda_1 = -3, \lambda_2 = -3, \lambda_3 = 5$$

$$A^3 + A^2 - 21A - 45I = 0 \Rightarrow A^3 + A^2 - 21A = 45I = \begin{bmatrix} 45 & 0 & 0 \\ 0 & 45 & 0 \\ 0 & 0 & 45 \end{bmatrix}$$

$$\text{Eigenvalues} = 5, -3, -3$$

$$\Rightarrow \text{Trace}(A) = \sum \lambda_i = \lambda_1 + \lambda_2 + \lambda_3 = 5 - 3 - 3 = -1$$

$$\text{And Determinant}(A) = \prod \lambda_i = \lambda_1 \times \lambda_2 \times \lambda_3 = (5)(-3)(-3) = 45$$

Eigenvector:

For $\lambda = 5$

$$[A - 5I]X_1 = \begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 & 2 & -3 \\ 1 & -2 & -3 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \left(R_2 \rightarrow \frac{1}{2}R_2 \right)$$

$$= \begin{bmatrix} -7 & 2 & -3 \\ 1 & -2 & -3 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \left(R_3 \rightarrow R_3 + R_2 \text{ \& } R_3 \rightarrow \left(\frac{-1}{4} \right) R_3 \right)$$

$$= \begin{bmatrix} 0 & 1 & 2 \\ 1 & -2 & -3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \left(R_1 \rightarrow R_1 + 7R_2, R_1 \rightarrow -\frac{R_1}{12} \right)$$

$$x_2 + 2x_3 = 0, x_1 - 2x_2 - 3x_3 = 0, x_2 = -2x_3$$

Let, $x_3 = k, x_2 = -2k$

Then $x_1 = 2x_2 + 3x_3 = -4k + 3k = -k$

$$X_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} \text{ or } k \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

So For $\lambda_1 = 5$ (when $k = 1$) $\Rightarrow X_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, Normalized form: $X_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

For degenerate eigenvalue, $\lambda_2 = -3$

$$|A - \lambda I| = 0 \Rightarrow \begin{bmatrix} -2 - \lambda & 2 & -3 \\ 2 & 1 - \lambda & -6 \\ -1 & -2 & 0 - \lambda \end{bmatrix} = 0$$

$$= \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & -3 \\ 1 & 2 & -3 \\ -1 & -2 & 3 \end{bmatrix}, \left(R_2 \rightarrow \frac{R_2}{2} \right)$$

$$= \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, (R_1 \rightarrow R_1 - R_2 \text{ \& } R_3 \rightarrow R_3 - R_2)$$

$$x_1 + 2x_2 - 3x_3 = 0$$

let $x_1 = k_1$ & $x_2 = k_2 \Rightarrow x_3 = \frac{k_1 + 2k_2}{3}$

$$X_2 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} k_1 \\ k_2 \\ \frac{1}{3}(k_1 + 2k_2) \end{bmatrix}$$

So For $\lambda_2 = -3$ (when $k_1 = k_2 = 1$) $\Rightarrow X_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, Normalized form: $X_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Taking X_3 such that X_2 and X_3 are linearly independent.

$$X_3 = \begin{bmatrix} 1 \\ 0 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$